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Project:- Smart Parking System

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1. Introduction

The **Smart Parking System** is a software-based system designed to make the process of finding, reserving, and managing parking spaces easier and more efficient. In traditional parking systems, drivers often spend considerable time searching for available parking spaces, which can result in traffic congestion, fuel consumption, and inconvenience.

The proposed system provides a centralized platform for managing parking spaces and allows users to check parking availability, reserve a suitable parking space, and manage their parking details. The system can also help parking administrators monitor parking spaces, manage bookings, and maintain parking-related information.

1.1 Purpose

The purpose of this SRS document is to define the complete requirements of the **Smart Parking System** in a clear and structured manner.

The system is intended to:

- Provide users with information about available and occupied parking spaces.
- Allow users to search for suitable parking spaces.
- Allow users to reserve a parking space.
- Reduce the time required to find available parking.
- Help administrators manage parking spaces and user bookings.
- Maintain accurate records of parking reservations and availability.
- Improve the overall efficiency and convenience of parking management.

1.2 Document Conventions

The following conventions are used throughout this SRS document:

- **Bold text** is used for important terms, headings, and system components.
- *Italic text* may be used for emphasis or specific terminology.

- Numbered headings are used to organize the document into chapters and sections.
- **FR** refers to a *Functional Requirement* of the system.
- **NFR** refers to a *Non-Functional Requirement* of the system.
- The terms **User**, **Customer**, and **Driver** refer to a person using the parking system to find or reserve a parking space.
- The term **Administrator** refers to the person responsible for managing parking spaces, users, and reservations.
- Requirements are written using clear and specific statements to avoid ambiguity.
- The word **"shall"** indicates a mandatory requirement of the system.

1.3 Intended Audience and Reading Suggestions

This SRS document is intended for everyone involved in the development, implementation, testing, management, and evaluation of the Smart Parking System.

The intended audience includes:

- **Project Developers:** To understand the required functionality and technical expectations of the system.
- **Project Guide/Supervisor:** To review the requirements and ensure that the project meets its objectives.
- **System Administrators:** To understand the features related to managing parking spaces, users, and reservations.
- **Testers:** To use the documented requirements as a basis for preparing test cases and verifying system functionality.
- **End Users/Drivers:** To understand the basic purpose and functionality of the parking system.
- **Project Evaluators:** To evaluate whether the developed system satisfies the specified requirements.

1.4 Product Scope

The **Smart Parking System** is intended to provide an efficient and convenient solution for managing parking spaces through a digital platform. The system will connect parking-space information with users so that available spaces can be identified and managed more effectively.

The major scope of the system includes:

- **User Registration and Login:** Users can create an account and securely log in to the system.
- **Parking Space Availability:** Users can view available and occupied parking spaces.
- **Parking Search:** Users can search for suitable parking spaces based on the available information.
- **Parking Reservation:** Users can reserve an available parking space.
- **Booking Management:** Users can view and manage their parking reservations.
- **Parking Management:** Administrators can add, update, remove, and manage parking spaces.
- **User Management:** Administrators can manage registered users and their parking activities.
- **Booking Records:** The system can maintain records of parking reservations.
- **Status Management:** Parking spaces can be represented as available, occupied, or reserved.
- **Improved Efficiency:** The system aims to reduce the time and effort required to find and manage parking spaces.

2. Overall Description

2.1 Product Perspective

The **Smart Parking System** is a standalone software system designed to improve traditional parking management. It provides a digital platform for users to view parking availability and make reservations, while administrators can manage parking spaces and bookings.

2.2 Product Functions

The main functions of the Smart Parking System are:

- User registration and login.
- Viewing available parking spaces.
- Searching for suitable parking spaces.
- Reserving a parking space.
- Viewing and managing bookings.
- Managing parking spaces by the administrator.
- Managing user and booking records.
- Updating parking space status.

2.3 User Classes and Characteristics

The system mainly has two types of users:

1. User/Driver:
Users can register, log in, search for available parking spaces, make reservations, and view their booking details. Basic computer or smartphone knowledge is sufficient.

2. Administrator:
The administrator manages parking spaces, users, reservations, and parking

status. The administrator should have basic knowledge of computer and system management.

2.4 Operating Environment

The Smart Parking System is designed to operate in a web-based environment.

- **Client:** Desktop, laptop, tablet, or smartphone.
- **Operating Systems:** Windows, Linux, Android, or other modern operating systems.
- **Browser:** Google Chrome, Microsoft Edge, Mozilla Firefox, or similar modern browsers.
- **Server:** A suitable web server or local development server.
- **Database:** A database system for storing user, parking, and booking information, if applicable.

2.5 Design and Implementation Constraints

The system may have the following constraints:

- The system requires an internet or network connection for online access.
- Users must have a compatible web browser.
- The system must provide a simple and user-friendly interface.
- Only authorized administrators should manage system records.
- The system must maintain accurate parking availability and booking information.
- The implementation must follow the technologies and resources available for the project.

2.6 User Documentation

The system will provide basic instructions to help users understand its functionality. The documentation will explain:

- How to register and log in.
- How to search for parking spaces.
- How to reserve a parking space.
- How to view and manage bookings.
- How administrators can manage parking spaces and records.

2.7 Assumptions and Dependencies

➢ Assumptions

- Users have basic knowledge of using websites.
- Users provide correct registration and booking information.
- Parking availability information is regularly updated.
- Administrators maintain accurate parking and user records.

➢ Dependencies

- The system depends on a working internet/network connection for online operation.
- The system depends on a compatible web browser.
- The system depends on the database for storing and retrieving system information.
- The system depends on proper server operation for accessing the application.

3. Functional Requirements

The functional requirements describe the main operations and services that the Smart Parking System shall provide to users and administrators. These requirements define how the system will perform its core functions.

3.1 User Registration and Login

FR1 – User Registration: The system shall allow new users to create an account.

FR2 – User Login: The system shall allow registered users to log in using valid credentials.

FR3 – Login Validation: The system shall display an error message for invalid login details.

3.2 User Profile Management

FR4 – View Profile: The system shall allow users to view their profile information.

FR5 – Update Profile: The system shall allow users to update their basic profile details.

3.3 Parking Space Search

FR6 – Search Parking: The system shall allow users to search for available parking spaces.

FR7 – View Parking Details: The system shall display the parking space number and its availability status.

3.4 Parking Availability Management

FR8 – View Availability: The system shall display the current status of parking spaces.

FR9 – Update Status: The system shall update the parking status when a space is reserved, occupied, or released.

3.5 Parking Reservation

FR10 – Select Parking Space: The system shall allow users to select an available parking space.

FR11 – Reserve Parking: The system shall allow users to reserve the selected parking space.

FR12 – Prevent Duplicate Booking: The system shall not allow a reserved or occupied space to be booked again.

3.6 Booking Management

FR13 – View Booking: The system shall allow users to view their booking details.

FR14 – Cancel Booking: The system shall allow users to cancel a booking, if applicable.

FR15 – Update Booking Status: The system shall update the booking status after cancellation or completion.

3.7 Parking Space Management

FR16 – Add Parking Space: The administrator shall be able to add new parking spaces.

FR17 – Update Parking Space: The administrator shall be able to update parking space information.

FR18 – Delete Parking Space: The administrator shall be able to remove parking spaces when required.

3.8 User Management

FR19 – View Users: The administrator shall be able to view registered users.

FR20 – Manage Users: The administrator shall be able to manage user accounts.

3.9 Booking Record Management

FR21 – Store Booking: The system shall store parking reservation records.

FR22 – View Booking Records: The administrator shall be able to view booking records.

3.10 Notifications and Messages

FR23 – Confirmation Message: The system shall display a confirmation message after successful operations.

FR24 – Error Message: The system shall display an appropriate message when an operation fails.

3.11 Logout

FR25 – User Logout: The system shall allow users and administrators to securely log out.

3.12 Access Control

FR26 – Role-Based Access: The system shall provide different access levels for users and administrators.

FR27 – Admin Security: The system shall restrict administrative functions to authorized administrators.

4. Non-Functional Requirements

Non-functional requirements define the **quality, performance, security, and usability** aspects of the Smart Parking System.

4.1 Performance

NFR1 – Response Time: The system shall respond to user requests within a reasonable time.

NFR2 – System Performance: The system shall efficiently handle multiple users and parking records.

4.2 Usability

NFR3 – User-Friendly Interface: The system shall provide a simple and easy-to-understand interface.

NFR4 – Easy Navigation: Users shall be able to navigate between system features easily.

4.3 Security

NFR5 – User Authentication: The system shall allow access only to users with valid login credentials.

NFR6 – Data Protection: The system shall protect user, booking, and parking information from unauthorized access.

4.4 Reliability

NFR7 – Accurate Information: The system shall provide accurate parking availability and booking information.

NFR8 – Error Handling: The system shall handle errors and display appropriate messages without crashing.

4.5 Availability

NFR9 – System Availability: The system shall be available whenever the

required server and network services are operational.

NFR10 – Data Availability: Stored user and booking information shall be available when required by authorized users.

4.6 Maintainability

NFR11 – Easy Maintenance: The system shall be designed so that its components can be easily maintained and updated.

NFR12 – Modular Design: The system shall organize major functions into manageable modules.

4.7 Scalability

NFR13 – Future Expansion: The system shall allow additional parking spaces, users, and features to be added in the future.

4.8 Compatibility

NFR14 – Browser Compatibility: The system shall work with commonly used modern web browsers.

NFR15 – Device Compatibility: The system shall be accessible on desktops, laptops, tablets, and smartphones.

5. Conclusion

The **Smart Parking System** provides a simple, efficient, and organized solution for managing parking spaces through a digital platform. It helps users search for available parking spaces, make reservations, and manage their bookings conveniently. At the same time, administrators can manage parking spaces, users, and booking records effectively.

The system is designed to reduce the time and effort required to find parking and minimize problems associated with traditional parking management. By providing accurate parking availability and reservation information, the system can improve the overall parking experience for users.

This SRS document defines the functional and non-functional requirements of the Smart Parking System. It provides a clear foundation for the system's design, development, testing, and implementation and ensures that the developed system meets the expected requirements.

The proposed system can also be enhanced in the future by adding features such as online payment, real-time location tracking, automated parking sensors, notifications, and vehicle number recognition. These improvements can make the system more advanced, reliable, and suitable for larger parking environments.

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